

Theory 13.4.1

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1 Equivariance and invariance

If $\mathbf{P}$ is a permutation matrix, then the following holds:

$$\mathbf{H}_{k+1}\mathbf{P} = \mathbf{F}[\mathbf{H}_k\mathbf{P}, \mathbf{P}^T\mathbf{A}\mathbf{P}, \phi_k].$$

2 Problem 13.7

Consider the simple GNN layer:

$$\mathbf{H}_{k+1} = \text{GraphLayer}[\mathbf{H}_k, \mathbf{A}] = \mathbf{a}[\boldsymbol{\beta}_k \mathbf{1}^T + \boldsymbol{\Omega}_k \begin{bmatrix} \mathbf{H}_k \\ \mathbf{H}_k \mathbf{A} \end{bmatrix}]$$